Siddharth Iyer

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Education

University of Toronto April 2028

Bachelor of Science (Honours) - Computer Science Specialist, Mathematics Minor

- Relevant Coursework: Data Structures and Algorithms, Databases, Machine Learning, Artificial Intelligence, Operating Systems, Web Development, Probability and Statistics, Theory of Computability.

Experience

Scotiabank | *Power BI, Azure, Linux, Git* Sept 2026 - Present

Cloud Engineer *Toronto, ON*

- Automating data preparation for collections dashboards to only 5 seconds with python scripts.
- Using SQL to explore and analyze collections activity data in an ETL pipeline spanning Azure and Linux servers.

Axiom Group Inc. | *Angular, .NET Core, SSMS, Node.js, Git, Jira* May 2025 - Aug 2025

Software Developer *Aurora, ON*

- Deployed and maintained ERP (Prism) features **used by 200+ employees** across Canada, USA, and Mexico.
- Streamlined Angular migration from v8 to v19, reducing live production build times **by 25+ seconds**.
- Collaborated with department leads and managers to **deliver 15+ production ERP features**.

Projects

BigVul Vulnerability Analyzer | *PyTorch, CodeBERT, scikit-learn, pandas, FastAPI, Git* GitHub

- Created a code safety tool using CodeBERT, fine-tuned with PyTorch, to classify C snippets as vulnerable.
- Fine-tuned CodeBERT **on 1,200 samples, achieving a 67% F1 score** for vulnerable-code classification.
- Reviewed the prototype for entrepreneurial scalability with ML engineers at LipDub AI, an AI video localization startup, for future migrations like AI tooling or an LLM agent skill for more convenient developer use.

Fly-Easy | *React Native, Flask, Google Firebase, Git* GitHub

- **Led a team of 5** people to create a mobile React Native, Flask, and Firebase airport companion app.
- Integrated Aerodatabox API for realtime airport/flight status, and Google's Places API for location tracking.
- Built greedy constraint-solving algorithm to generate an event planner with user preferences and scheduling.
- Assessed efficiency of greedy algorithm using evaluation scripts against Claude Sonnet 4.5's constraint solving, receiving an average score of **83.4 versus Claude's 83**, due to **20% better hard constraint satisfaction**.

CourtClash | *React, Node.js, Express, PostgreSQL, Socket.IO, Stripe, OAuth2* GitHub

- Engineered a debate platform with React and Express, culminated with an AI-generated verdict.
- Integrated Google OAuth2 and Stripe subscriptions to authenticate users and control access to application.
- Orchestrated a **2-stage AI workflow** combining argument analysis with transcript-based debate judging.
- Maintained room history with pagination, reducing response sizes by **99% against a 1,000-record baseline.**

Pathway | *Next.js, OpenAI, Git* GitHub

- Developed a RAG-style course planning assistant using Next.js, OpenAI API, and ChromaDB.
- Improved efficiency by **reducing course search time by an estimated 70%** compared to manual browsing.
- Utilized Python web-scraping library, BeautifulSoup, on a university website for course info, prereqs, and timings.

mysh.c | *C, TCP/IP, Makefile, Bash, Linux/Unix* GitHub

- Engineered a lightweight Unix-like shell in C supporting process control, piping, and job management.
- Implemented TCP socket-based networking for remote execution in inter-process, inter-system communication.
- Designed robust signal handling to manage concurrent processes and prevent zombie/orphan processes.

Technical Skills

Languages: Python, Java, C, C++, HTML/CSS, JavaScript, TypeScript, SQL, C#, Bash, Assembly
Frameworks/Libraries: Angular, Next.js, React, React Native, .NET Core, Flask, NumPy, PyTorch, scikit-learn
Databases: Microsoft SQL Server, PostgreSQL, MySQL
Cloud & Tools: Azure, Firebase, Git, Docker, Node.js, Linux, Jira, PowerBI